From Archive to Interface: Positioning the Digital Catalogue Raisonné within Nordic Digital Humanities Infrastructure

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Abstract

This article examines the digital catalogue raisonné as a critical intersection of infrastructure and interface within Nordic digital humanities. Drawing on the art scholarly database as conceptualized by Bearman (1988) and contemporary Digital Humanities frameworks (Drucker 2021), this research positions the digital catalogue raisonné as a specialized scholarly resource functioning within broader digital humanities infrastructure. Implementing interdisciplinary approaches that integrate design methods, database management, accessibility frameworks, and sustainability planning, this research positions the digital catalogue raisonné as a scholarly publication. Using comparative analysis of the Nordic catalogue raisonné projects on Hilma af Klint, Akseli Gallen-Kallela, Ernest Mancoba, and Edvard Munch, and emerging initiatives on Torsten Andersson and Vera Nilsson, this text shows the capability of design-centered methodological frameworks to advance accessibility and sustainability in digital art historical documentation. The article contributes to ongoing discussions within Nordic digital humanities about user-centered design, open access principles, collaborative scholarship, and the democratization of cultural heritage resources.

Keywords

Digital catalogue raisonné, Nordic digital humanities, infrastructure, interface design, accessibility, sustainability, Hilma af Klint, Edvard Munch, cultural heritage

1. Introduction

A catalogue raisonné is, by definition, an impossible project. It promises to document every work by an artist: a task that requires locating the unlocated, attributing the disputed, and preserving the ephemeral, all while maintaining scholarly authority over a record that will inevitably need updating. For centuries, print made this impossibility manageable by making it final. Digital platforms have removed that excuse. The question now is not whether catalogues raisonnés should evolve, but how to build the infrastructure and interfaces that let them do so responsibly.

This article was written for the Norwegian Network for Digital Humanities and Cultural Organization (DHKO) 2025 conference at the University of Inland Norway in Hamar, whose theme "Infrastructure and Interface" provides a lens for the digital catalogue raisonné. The genre sits precisely at that intersection: it is both infrastructure for art historical scholarship and an interface between archival materials and scholarly engagement. This article extends that positioning by analyzing how design-centered methodologies advance infrastructure development and user accessibility.

The digital catalogue raisonné represents a convergence of traditional art historical scholarship and contemporary digital humanities (DH) methodologies, creating new possibilities for accessibility, collaboration, and sustainable knowledge preservation (Gabrielli 2015; Friedenthal 2020). As Drucker (2021) argues, DH infrastructure supports cultural scholarship through computational methodologies and democratization of scholarly resources. A catalogue raisonné, which is defined as an authoritative, annotated compilation of all known works by an artist, provides provenance documentation, scholarly annotation, and comprehensive information that supports academic research and the preservation of cultural heritage (Franklin 2012; Friedenthal 2020).

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This article explores how design-centered methodological frameworks advance accessibility and sustainability in Nordic digital catalogue raisonné projects. The text also examines infrastructure models that best support the development of these specialized scholarly resources within the broader Nordic DH ecosystem. Design-centered, regionally coordinated infrastructure, rather than isolated institutional projects is the most effective and sustainable model for Nordic digital catalogue raisonné development.

Rooted in Bearman's (1988) concept of the art scholarly database, contemporary digital catalogues raisonnés extend these foundational principles to develop advanced information systems supporting art historical research and preservation. Within the Nordic context, digital catalogue raisonné projects have gained momentum through major initiatives documenting Hilma af Klint, Akseli Gallen-Kallela, Ernest Mancoba, and Edvard Munch and emerging projects on Torsten Andersson's paintings and Vera Nilsson's drawings. These projects show how Nordic institutions are developing infrastructure, such as technological platforms, metadata standards, and collaborative frameworks, that supports digital scholarship and the interfaces (design systems, user experiences, and accessibility considerations) that enable increased engagement with cultural heritage resources.

This article contributes to the DHNB (Digital Humanities in the Nordic and Baltic Countries) community by applying design-centered frameworks to catalogue raisonné development, demonstrating how specialized scholarly resources can advance DH research, education, and communication across Nordic and Baltic countries. Drawing on research through design methodology from Fox-Jensen's (2025) research, developed further in Fox-Jensen (2026a) and Fox-Jensen (forthcoming), this analysis shows how interdisciplinary approaches integrating design methods, DH, and participatory research can simultaneously advance infrastructure and interface development. The case study analysis draws on publicly available project documentation, including institutional websites, published digital interfaces, and online project descriptions, supplemented by Fox-Jensen's direct examination of each platform. Structuring the transformation from "Archive to Interface" (Figure 1) illustrates how design thinking, collaborative methods, and sustainability strategies can improve accessibility in digital catalogue raisonné projects while establishing frameworks applicable across diverse artistic oeuvres and institutional contexts. This framework highlights the need for coordinated efforts supporting ongoing and future projects across the Nordic region. Nordic Art Intelligence foundation established in 2021 coordinates catalogue raisonné research and resources throughout the region.

This article further responds to that need through four interconnected sections. Section 1 provides an introduction and establishes scholarly context, and section 2 establishes theoretical frameworks integrating textual scholarship, biographical research, digital scholarly editing, and DH principles applicable to catalogue raisonné development within the Nordic context. Section 3 applies these frameworks through comparative analysis of major Nordic catalogue raisonné projects, examining diverse institutional models, publication formats, design methods, and sustainability strategies. Section 4 synthesizes key findings and articulates future directions for regional coordination, offering practical recommendations for Nordic DH infrastructures. Together, these sections show how catalogue raisonné projects function as specialized scholarly publications and address challenges of digital infrastructures.

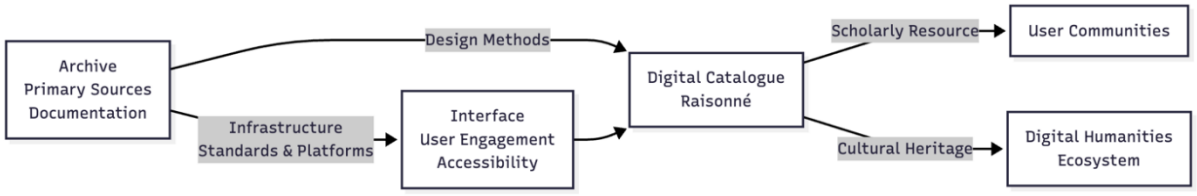


Figure 1: "Archive to Interface" framework illustrating relationship between archival infrastructure and user engagement. Created by Fox-Jensen with Mermaid AI, 2025.

2. Theoretical Frameworks and Methodological Approaches

The digital catalogue raisonné operates at the intersection of textual scholarship, DH, and design-centered research, functioning simultaneously as scholarly publication and DH infrastructure. Informed by Bearman's (1988) art scholarly database framework, this text positions catalogues raisonnés as evolving systems that combine structured information organization with interpretive scholarship. This interdisciplinary positioning requires integration of bibliographical methods, biographical theory, and contemporary digital scholarly editing practices.

2.1. Textual Scholarship and Bibliographical Foundations

Textual scholarship provides fundamental frameworks for catalogue raisonné projects. Bibliographical theory, applied to art historical documentation, underpins this approach (Friedenthal 2012; Fox-Jensen 2025). The catalogue raisonné extends bibliographical methods further than traditional textual documentation to systematic cataloguing of artworks by treating each piece as an entry within a rigorously organized scholarly publication (Egan and Shera 1952; Greetham 1994). This approach documents physical properties, provenance histories, and contextual interpretations, ensuring each artwork receives detailed scholarly attention.

Macrocosmic and microcosmic textual perspectives enable catalogue raisonné scholarship to contextualize artworks within broad historical frameworks while examining intrinsic details of individual works (McGann 2001). Bryant's (2002) fluid text theory proves particularly relevant, acknowledging how catalogue entries evolve as new research emerges and supporting versioning. In addition to transcription, editorial interventions generate new interpretive layers, what Ohge (2021) terms "enjoyed meanings," thereby enriching the scholarly apparatus through curatorial decisions and annotations. This recognition of catalogues as "living documents" supports iterative development and versioning practices to sustain long-term publications (Fox-Jensen 2025, 2026a), particularly evident for Nordic projects with digital platforms that enable ongoing updates, as seen in the Munch (2025) and Gallen-Kallela catalogues raisonnés.

2.2. Biography and Archival Documentation

Biographical research forms another foundational component of catalogue raisonné scholarship, informing the relationships between artists' lived experiences and creative output (Honan 1979). Archival materials (correspondence, personal journals, and ephemera) provide primary source evidence to support catalogue raisonné textual components (Salvesen and Cousineau 2005; Würtenberger 2016). The integration of biographical essays within catalogue raisonné publications advances traditional chronological structures to situate artists within broader social, cultural, and historical contexts (Hemecker and Saunders 2017), as in the major Nordic catalogue raisonné projects documenting Munch, af Klint, and Gallen-Kallela.

2.3. Digital Scholarly Editing and Collaborative Approaches

Digital scholarly editing methodologies for catalogues raisonnés research through collaborative approaches, improved accessibility, and adaptable publication formats (Pierazzo 2015). Digital editing practices enable projects to incorporate annotations, commentary, and collaborative contributions extending scholarly dialogue beyond traditional print limitations (Brown 2011; Kaltenbrunner 2015). These approaches facilitate dynamic updating of catalogue entries, integration of multimedia documentation, and implementation of user-centered interface design to enhance research access.

I argue that digital catalogues raisonnés must uphold the editorial standards established by traditional print scholarship. Following Drucker (2021) on DH infrastructure and St-Pierre (2022) on the complexity of the scholarly apparatus, this means actively embracing, rather than merely accommodating, the interactivity, searchability, and ongoing revisability that digital platforms afford. This dual commitment ensures projects maintain scholarly authority while increasing accessibility

across diverse user communities, a balance particularly important within Nordic DH where open access (OA) principles align with regional commitments to democratized cultural heritage resources and projects.

2.4. Positioning within Nordic Digital Humanities Infrastructure

DH frameworks position the catalogue raisonné as infrastructure that supports art historical and cultural scholarship through computational methodologies, interdisciplinary collaboration, and democratization of scholarly resources (Drucker 2021). The field's emphasis on method and interdisciplinarity aligns with catalogue raisonné development, which integrates expertise from art history, cultural resources, design, database management, metadata standards, and information visualization (Repko 2020). Drucker's (2021) conception of DH as fundamentally concerned with the interpretive dimensions of digital tools and platforms emphasizes how catalogues raisonnés function as databases and interpretive scholarly publications.

Nordic projects that exemplify this commitment are the Munch's open digital catalogue, the Gallen-Kallela project's OA coordination through Nordic Art Intelligence (2021) foundation and emerging initiatives that prioritize accessibility alongside scholarly rigor. However, democratization operates within constraints imposed by authentication requirements and scholarly validation processes, within which subject-matter experts maintain authority over attribution decisions and inclusion criteria. Design approaches enhance inclusivity through accessible interfaces, comprehensive metadata, and attention to diverse user needs across varied contexts of use.

This theoretical and methodological foundation informs the subsequent analysis of Nordic catalogue raisonné projects, demonstrating how interdisciplinary approaches that integrate textual scholarship, biographical research, digital editing practices, and DH principles advance scholarly rigor and practical accessibility in contemporary art historical documentation.

3. Design Methods for Nordic Catalogues Raisonnés

This section examines the design methodologies and institutional approaches seen across major Nordic catalogue raisonné projects, analyzing how interdisciplinary frameworks integrating user-centered design, accessibility considerations, and sustainability planning advance scholarly rigor while expanding public engagement with cultural heritage resources.

3.1. Edvard Munch: Infrastructure, Scale, and User-Centered Design

The Edvard Munch (1863–1944) catalogue raisonné represents one of the most comprehensive Nordic digital art documentation projects, illustrating how large-scale infrastructure can support sustained scholarly engagement and serve diverse user communities locally and internationally. The complete paintings catalogue, published in print in 2008 under Gerd Woll's (2008) editorship, comprises four volumes totaling 2,200 pages and documenting 1,789 paintings through collaboration between the Munch, the Faurschou Foundation, and Kaare Berntsen.

The Munch (2025) digital catalogue raisonné includes four volumes based on different media: 1. paintings (published 2008, originally print, now digital), 2. graphic works and prints (published 2001, print), 3. writings (published 2010, digital), and 4. drawings (published 2013, digital). This phased volume development reflects the museum's strategic approach to digitization, recognizing that the print catalogues for paintings and prints totaled 16 kilograms (30 pounds), a hefty weight that limited accessibility despite its comprehensive documentation. Digital formats improved accessibility, searchability, and ongoing versioning. These updates may take the form of additional provenance, exhibition history and photographs, condition reports, and bibliographic textual documentation.

The digital paintings catalogue documents each work with detailed catalogue entries that include reference number, images, title, date, dimensions, medium, technique, provenance, exhibition history, and bibliographic references. High-resolution images enable close examination of details of brushwork, surface texture, and condition impossible in print reproduction. The systematic documentation supports multiple research approaches, from iconographic studies that trace motif and series development across

decades to technical analyses that examine Munch's evolving material practices and provide additional historic context.

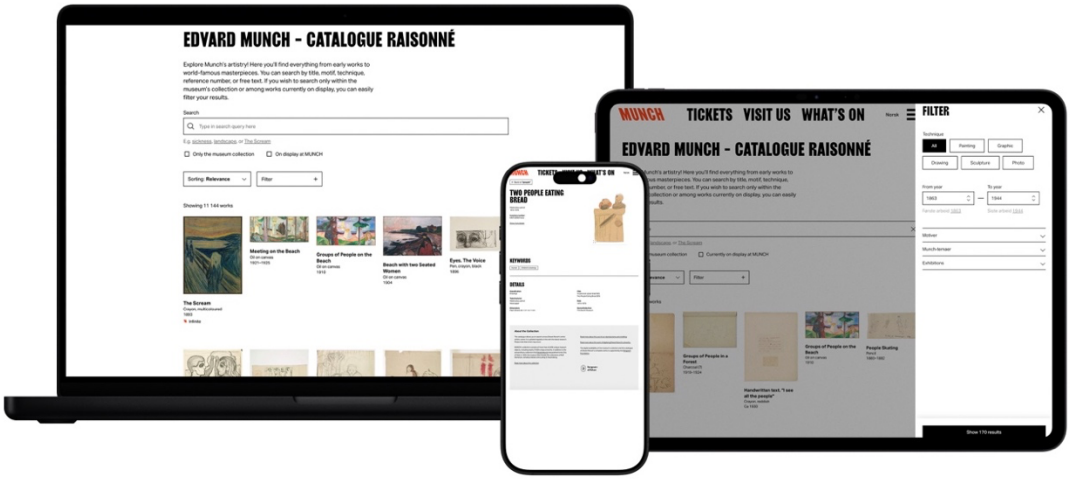


Figure 2: Screenshots of the *Munch Catalogue Raisonné* digital online interface displayed across responsive screen sizes, including desktop, tablet, and mobile. The digital publication is hosted within the Munch website and houses 11,145 objects from the various previous catalogue raisonné volumes.

3.1.1. Design Methods in Practice

Drawing on Holtzblatt and Beyer's (2022) contextual design principles, I interpret the Munch catalogue as a model for large-scale digital publication, one that succeeds by attending systematically to user needs across varied contexts of use. The comprehensive yet simple interface enables users to search by title, motif, technique, reference number, or free text, with filtering options for works in the museum's collection in storage or currently on display. Advanced search functionality supports Boolean operators (words such as "AND," "OR," and "NOT," that allow you to refine your search parameters when using a search engine), date ranges, and combinatorial queries enabling complex research questions. For example, scholars can locate all self-portraits from 1880–90, all works using specific techniques, or all paintings currently accessible for study in Oslo.

This searchable system serves diverse user groups with different information needs. Munch scholars require comprehensive documentation supporting attribution decisions and provenance research. Museum professionals need exhibition history and condition information for loan requests and conservation planning. Students and general audiences seek accessible entry points to Munch's oeuvre through thematic browsing and visual exploration. Collectors and dealers require authentication information and market history documentation. The interface supports these varied needs through flexible search pathways and layered information architecture, allowing casual browsing while supporting specialized scholarly investigation.

The interface applies visual representation and design principles through consistent iconography in navigation, high-resolution zoomable images that support detailed examination (Google Material Design n.d.), and hierarchical organization of information (Blair-Early and Zender 2008). The modular responsive grid system adapts across platforms, from desktop computers to tablet devices to smartphones (Figure 2), maintaining functionality and readability across screen sizes. While the mobile version functions adequately, opportunities exist for optimization, particularly regarding the device breakpoint strategy that affects the scale of all the components negatively depending on the user and age of device.

My review of the Munch's published documentation indicates that user testing conducted during interface development revealed that art historians prioritize comprehensive search functionality and detailed metadata, while general audiences value visual browsing and contextual information that provides Munch's biographical and historical period context. The interface design balances these

priorities through dual navigation pathways, including systematic searching for research users and thematic browsing for exploratory users. The featured content on the catalogue raisonné homepage rotates regularly, highlighting particular works, recent scholarship, and connections to current exhibitions, maintaining engagement for returning users and new visitors.

3.1.2. Sustainability and Accessibility

The Munch project illustrates the four-dimensional sustainability framework (Figure 3) through institutional commitment to ongoing technical maintenance, economic stability via museum operating budgets, and organizational continuity through dedicated digital teams (Edmond and Morselli 2020; Fox-Jensen 2025, 2026a). The Munch's commitment to ongoing maintenance ensures the digital catalogue remains functional as web standards evolve and user expectations change.

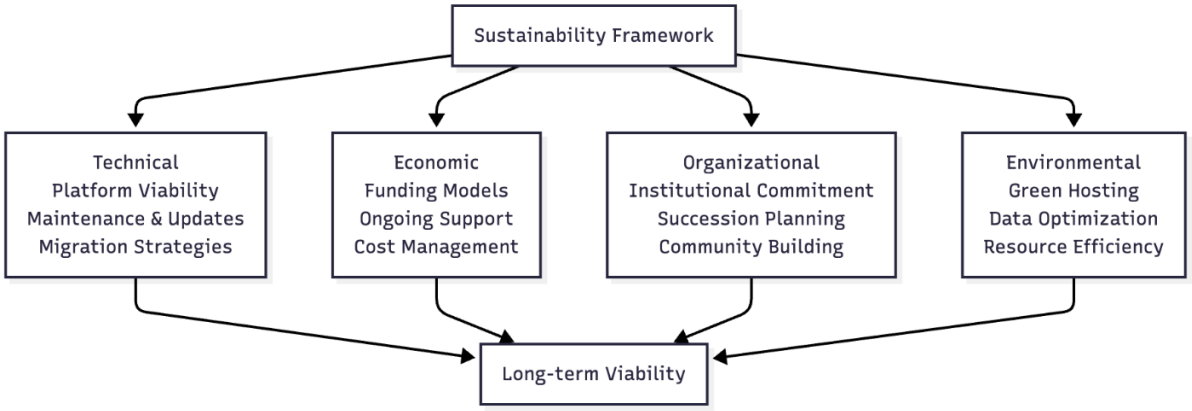


Figure 3: "Sustainability framework" for digital catalogue raisonné projects organized by technical, economic, organizational (collaborative), and environmental. Created by Fox-Jensen with Mermaid AI, 2025.

Economic sustainability involves funding models that support initial development and ongoing maintenance. The Munch catalogue benefited from institutional investment, which reflects the museum's recognition that Munch documentation constitutes core mission activity. Annual budgets support hosting costs, security updates, content additions as new works are documented and attributed, and periodic interface improvements responding to user feedback. This sustained funding commitment distinguishes institutional projects from grant-dependent initiatives that are vulnerable to discontinuation when external funding expires.

Organizational sustainability depends on institutional commitment, staff expertise, and governance structures that ensure continuity despite personnel changes. Based on published institutional documentation, the Munch's digital team includes database administrators, web developers, and collection managers who collaboratively maintain the catalogue. Documentation of system architecture, content standards, and workflow procedures enables knowledge transfer as staff transition, preventing loss of institutional memory. Regular stakeholder meetings involving curators, educators, and IT staff maintain alignment in the technical infrastructure and scholarship.

Environmental sustainability increasingly relates to DH projects, as data centers significantly consume energy and contribute to carbon emissions (Golin and Pearson 2022). The Munch may address environmental impact by choosing green hosting providers that use renewable energy, and efficient caching strategies that minimize redundant data transfers. While these measures represent initial steps, systematic environmental impact assessment remains an emerging priority for digital cultural heritage projects throughout the Nordic region.

The Munch's OA approach aligns with Nordic DH trends toward democratizing cultural heritage resources, supporting broader European commitments articulated through European and national digitization strategies (European Commission n.d.). Free, unrestricted access removes economic barriers to engagement, supporting educational uses, independent scholarship, and creative

appropriations that extend Munch's cultural influence. Download options enable users to incorporate high-resolution images in presentations, publications, and educational materials, with clear copyright licensing information facilitating legal reuse.

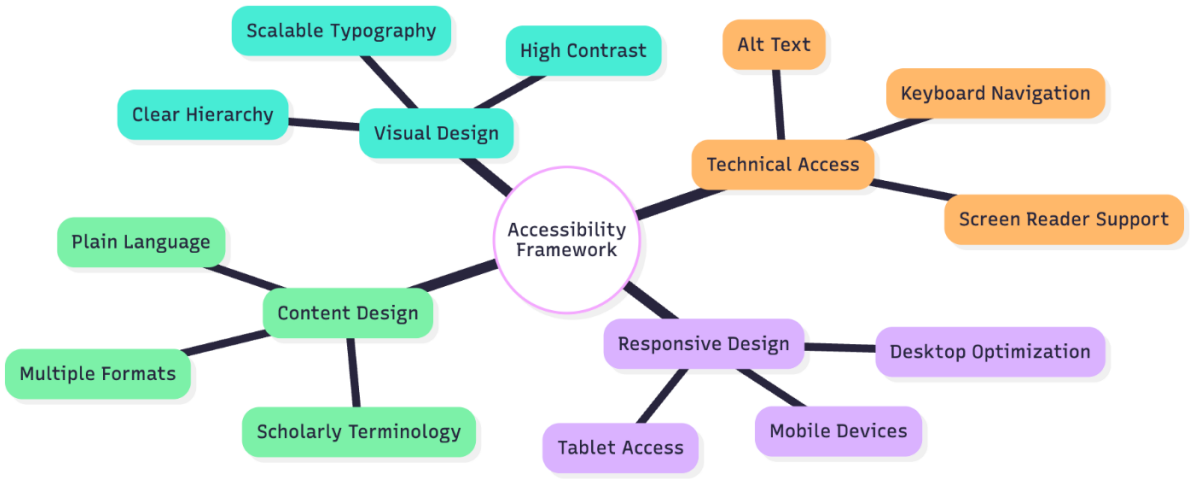


Figure 4: "Accessibility framework" for digital catalogue raisonné interfaces, organized by visual design, technical access, responsive design, and content. Created by Fox-Jensen with Mermaid AI, 2025.

The accessibility framework (Figure 4) organizes considerations around visual design, technical access, responsive design, and content following Web Content Accessibility Guidelines (WCAG) (W3C 2018). Visual design accessibility includes sufficient color contrast for readability, scalable typography that supports user-controlled text sizing, and clear information hierarchy. Technical accessibility ensures effective keyboard navigation for users unable to use pointing devices, screen reader compatibility for visually impaired users, and alternative text descriptions for images. Responsive design adapts layouts across device types and screen sizes, maintaining functionality from large desktop monitors to small smartphone screens. The Munch catalogue raisonné exemplifies these tenets through its use of bold typefaces and generous white space, which improve the visual clarity and guide users through content hierarchies in any viewing context. Content accessibility provides descriptive metadata that enables multiple pathways to discovery, multilingual support serving international audiences, and clear, jargon-free language in contextual information.

The accessibility review I conducted on the live Munch catalogue raisonné interface identified areas for improvement, including inconsistent alternative (alt) text for decorative interface elements, keyboard focus indicators requiring enhanced visibility, and search result presentations challenging for screen reader users. The mobile version of the Munch catalogue raisonné particularly requires improved optimization to ensure touch-based navigation and responsive design adequately support users with disabilities across varied devices and screen sizes. Future remediation must address these accessibility considerations through iterative development cycles, establishing commitment to inclusive access as an ongoing priority rather than treating accessibility as a one-time compliance requirement.

3.2. Hilma af Klint: Scholarly Authority and Multiformat Publication

The Hilma af Klint (1862–1944) Catalogue Raisonné, published in seven volumes between 2020 and 2022, represents the first comprehensive catalogue raisonné for a Swedish female artist, documenting nearly 1,600 works (Birnbaum and Almqvist 2020–22). This monumental project has fundamentally transformed scholarly and public understanding of af Klint's significance as an abstract art pioneer, establishing definitive documentation supporting her recognition as among the earliest abstract painters, preceding the Russian artist Wassily Kandinsky's (1866–1944) commonly cited 1910 abstractions.

The timing of the project's completion proved fortuitous, coinciding with unprecedented international interest sparked by the landmark Guggenheim Museum presentation (2018–19), which attracted over 600,000 visitors, becoming the museum's most-attended exhibition. This visibility

amplified scholarly attention and market interest, supporting the catalogue raisonné's importance as authoritative reference for attribution, authentication, and contextual understanding. This symbiotic relationship shows how commercial viability and scholarly rigor can reinforce one another.

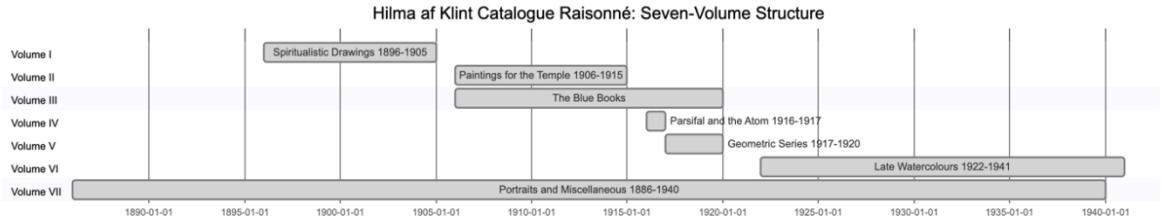


Figure 5: Hilma af Klint's *Catalogue Raisonné* seven-volume chronological organization of the volumes (Birnbaum and Almqvist 2020–22). Created by Fox-Jensen with Mermaid AI, 2025.

The seven-volume structure (Figure 5) reflects af Klint's own organizational principles and distinct creative phases established by Olof Sundström, the Swedish cataloguer/archivist who labored on cataloguing her work following Klint's death in 1944: *Spiritualistic Drawings, 1896–1905* (Volume I), *Paintings for the Temple, 1906–1915* (Volumes II–III), *The Blue Books* (Volume IV), *Parsifal and the Atom, 1916–1917* (Volume V), *Geometric Series, 1917–1920* (Volume VI), and *Portraits and Miscellaneous Works, 1886–1940*, plus *Late Watercolours, 1922–1941* (Volume VII). This chronological and thematic structure enables readers to trace af Klint's stylistic evolution, conceptual development, and sustained engagement with spiritual and scientific themes across five decades.

Each volume opens with scholarly essays that provide historical context, biographical information, and interpretive frameworks for understanding the documented works. Volume II, for example, includes extensive analysis of the *Paintings for the Temple* series, af Klint's most ambitious project comprising 193 paintings created in 1906–15 and made according to directives she believed came from spiritual guides. Essays examine af Klint's involvement with Theosophy and spiritualism, her relationship with Rudolf Steiner, and connections between her spiritual beliefs and abstract visual language. This contextual apparatus transforms the catalogue raisonné from reference work into interpretive publication, supporting scholarly engagement while maintaining documentary rigor.



Figure 6: *Hilma af Klint: Catalogue Raisonné* seven printed volumes (Birnbaum & Almqvist 2020–22).

3.2.1. Cataloguing Methodology and Scholarly Apparatus

The catalogue entries follow consistent format across all volumes, documenting each work with Swedish and English titles, creation date, dimensions, medium, technique, current location, provenance, exhibition history, and bibliographic references. Condition notes document conservation interventions and inform preservation planning. Many entries include af Klint's own annotations from notebooks and sketchbooks, providing primary source evidence of her conceptual intentions and working processes.

Attribution decisions reflect collaborative scholarly judgment of the editors Daniel Birnbaum and Kurt Almqvist, with advisory board members including international af Klint specialists. Works of uncertain attribution receive qualified designations with explanatory notes that discuss attribution evidence. This transparent methodology builds scholarly trust, acknowledging interpretation's role while maintaining authoritative standards. Several previously unknown works emerged during research, which are documented in appendices as pending authentication confirmation and verification.

The cataloguing process reconciled af Klint's documentation with physical works and documented damaged or lost works known only through notebooks or photographs. Approximately twenty paintings documented in af Klint's records remain unlocated but are included in the catalogue raisonné with documentation made available to facilitate future identification. This inclusive approach contrasts with catalogues that omit unlocated works, reflecting the editors' commitment to comprehensive documentation even when works cannot be physically examined. These resource-intensive practices, locating, verifying, and physically examining artworks, represent financial and temporal investments that substantially increase the timeline and budget required for catalogue raisonné development.

3.2.2. Design and Visual Representation

The seven-volume print publication features high-quality full-color reproductions with large-format pages that accommodate detail views and fold-out pages for monumental works.

Layout design integrates catalogue entries with visual materials, placing images adjacent to corresponding text rather than segregating illustrations in separate sections. This integrated approach facilitates comparative viewing and supports scholarly examination of series developments and recurring motifs. Fold-out pages accommodate af Klint's largest paintings, including the Group I, Primordial Chaos series paintings, which measure up to three meters in height.

Typography choices emphasize readability in extended scholarly apparatus, using generous margins, clear hierarchies to distinguish entry components, and consistent formatting across volumes. The physical volumes demonstrate attention to material quality, including durable binding, acid-free paper for longevity, and protective slipcases for the complete seven-volume set. These production values reflect commitment to creating lasting reference works worthy of af Klint's significance.

Aside from the traditional print publication, the foundation has explored immersive formats, including a digital publication with virtual and augmented reality experiences enabling viewers to encounter af Klint's monumental Paintings for the Temple at scale. These VR (virtual reality) and AR (augmented reality) implementations recognize that physical access to works remains limited: many paintings reside in the foundation collection with restricted public access, while others belong to international institutions requiring travel. Digital immersive experiences democratize access while creating novel engagement modes impossible in physical exhibition contexts. However, the digital catalogue raisonné and VR/AR experiences currently remain restricted, available only through institutional exhibitions rather than as OA resource.

While currently distributed through the publisher Bokförlaget Stolpe in multiple formats, the Hilma af Klint: Catalogue Raisonné could have its broader scholarly access enhanced by online open-access availability like that of the Munch catalogue, democratizing access for global researchers particularly in regions where purchase costs present barriers. As of 2026, no digital OA version of the catalogue raisonné exists online. Based on publicly available information from the foundation and Bokförlaget Stolpe, commercial digital formats including VR, AR, and NFTs have been prioritized over OA, potentially shaped by publisher contracts, revenue dependencies, and the technical demands of hosting a publication of this scale.

I propose that a viable model could position the catalogue raisonné within Swedish museum infrastructure parallel to the Munch's hosting of the Munch catalogue raisonné, or through partnership with the Moderna Museet (2024), which is slated to host the Vera Nilsson drawings catalogue raisonné.

3.2.3. Institutional Stewardship and Collaborative Infrastructure

The survival of nearly all works within the Hilma af Klint Foundation (2022) enabled comprehensive documentation, illustrating how institutional stewardship impacts catalogue raisonné feasibility. Af

Klint stipulated that the works remain unsold for twenty years following her death in 1944, believing audiences in the future would better understand her revolutionary abstractions. Her nephew Erik af Klint honored this directive, maintaining the collection intact, eventually transferring it to the foundation that was established 1972. This sustained family stewardship preserved the physical works and archival documentation, such as notebooks, sketches, correspondence, and af Klint's own cataloguing systems, providing the documentary foundation essential for authoritative catalogue raisonné production.

Af Klint's posthumous legacy, however, has become increasingly complex in recent years. Recent scholarship has raised questions about authorship, collaboration, and the narratives constructed around her work. Gregory (2025) documents that Kurt Almqvist, as he examined af Klint's 26,000 pages of journals, reveals the artist's personality as markedly different from the martyrlike portrait created by her nephew Erik af Klint in 1967. Also noteworthy, the historian Hedvig Martin's research, which cross-references journal entries with artworks, found evidence that the *Paintings for the Temple*, af Klint's magnum opus, was collaborative rather than the work of a single artist; Anna Cassel contributed approximately fourteen paintings to the *Primordial Chaos* series, as compared to af Klint's twelve.

The foundation's systematic preservation provided the basis for the catalogue raisonné and supported major international exhibitions at Moderna Museet Stockholm (2013), Solomon R. Guggenheim Museum New York (2018–19) and Guggenheim Bilbao (2024–25), and will support forthcoming presentations. Each exhibition generates scholarly research that is incorporated into ongoing af Klint documentation, showing how catalogue raisonné publication initiates rather than concludes scholarly engagement.

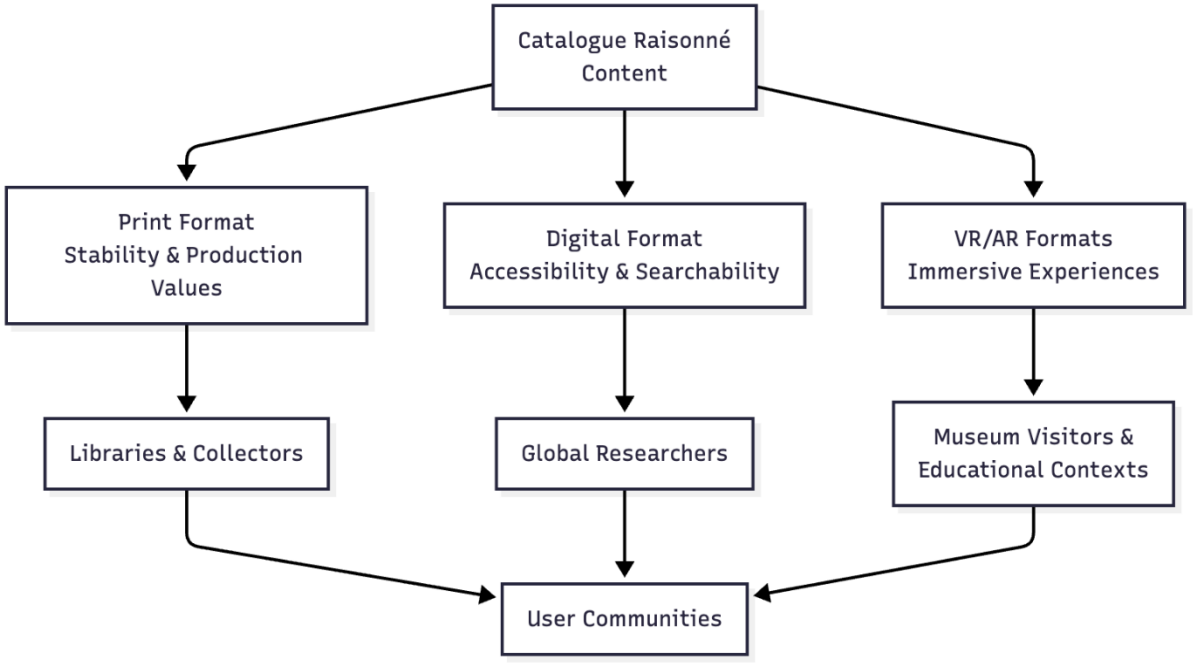


Figure 7: "Catalogue Raisonné Content" showing multiformat publications, including print, digital, and VR/AR formats, all facilitating access to various user communities and GLAMs. Created by Fox-Jensen with Mermaid AI, 2025.

The multistakeholder collaboration between the foundation, the museums (Guggenheim), the Axel and Margaret Ax:son Johnson Foundation (major funding partner), Bokförlaget Stolpe (publisher), and international scholars embodies participatory methods creating "trading zones" (Lauder and Salmon 2014) in which experts exchange ideas and negotiate shared understandings. Editorial meetings convened specialists in early abstraction, Swedish modernism, Theosophy and spiritualism, conservation science, and archival research, synthesizing diverse expertise into coherent documentation. This collaborative model often distributed intellectual authority while maintaining editorial coherence, balancing individual scholarly perspectives with collective judgment. Af Klint's

legacy has nonetheless been recently complicated by disputes between her heirs and the foundation board over commercialization, authentication, and public access to her posthumous works (Small 2023; Artnet 2025; Beeson 2025).

The multiformat approach (Figure 7) reflects democratization principles central to DH, giving access to different communities through print, digital, and immersive formats. Print volumes serve scholars, institutional libraries, and collectors valuing tangible reference works and permanent access independent of digital infrastructure. Digital formats (planned) would enable searchability, dynamic updates incorporating new research, and global access removing economic and geographic barriers. Immersive VR/AR experiences engage audiences through experiential encounters impossible in traditional formats, creating affective connections that complement scholarly documentation. This diversity of format recognizes that different modes serve different purposes, as comprehensive documentation requires multiple access pathways rather than single authoritative format.

3.3. Akseli Gallen-Kallela: Open Access and Collaborative Infrastructure

The Akseli Gallen-Kallela (1865–1931) catalogue raisonné project, managed by Nordic Art Intelligence foundation and launched digitally in February 2024 at gallenkallela.org, exemplifies contemporary OA approaches to art historical documentation. The project aims to document all known paintings by one of Finland's most significant painters from the fin-de-siècle period, who is known for National Romantic themes drawn from Finnish mythology, landscape, and folklore. His major works include the Kalevala series illustrations, such as the Aino Myth triptych, and portraits that integrate symbolist aesthetics with Finnish cultural nationalism.

The project builds upon forty years of research by the artist's granddaughter Aivi Gallen-Kallela-Sirén, whose lifelong dedication to documenting her grandfather's oeuvre provided foundational scholarship. Her comprehensive card index of the artist's painting oeuvre compiles exhibition histories, tracked provenance changes, identified works in private collections, and preserved family archival materials that document Gallen-Kallela's working methods and conceptual intentions. This sustained family scholarship represents invaluable intellectual capital that catalogue raisonné projects typically accumulate through institutional support.

Project researcher Irene Riihimäki organized, updated, cross-checked, and digitized the information from the card index for the Navigating.art (n.d.) platform. The core of the publication's bibliography and expography is based on Gallen-Kallela's exhibition history as provided by *Suomalaisen kuvataiteen bibliografia* (Bibliography of Finnish Art, n.d.), a service maintained by the Finnish National Gallery's Library. Riihimäki further expanded and updated the exhibition history and linked the exhibition data to individual paintings.

Coordination from Finnish institutions including the Ateneum Art Museum/Finnish National Gallery and Serlachius Museums (both holding a major Gallen-Kallela collections), and National Library of Finland (preserving extensive archival documentation) provided institutional expertise, collection access, and technical infrastructure. Nordic Art Intelligence foundation role as coordinating organization supplied digital platform development, project management, metadata standards, and connection to broader Nordic infrastructure initiatives.

3.3.1. Open Access and Regional Coordination

The project's OA commitment reflects contemporary trends toward democratizing art historical resources, aligning with European Commission (n.d.) open science policies and Nordic cultural values emphasizing public access to cultural heritage. Complete catalogue content appears freely accessible online without registration requirements or paywalls, enabling engagement by the global scholarly community regardless of institutional affiliation or geographic location. This access model fundamentally expands potential user communities, in contrast to subscription-based art databases that restrict access to paying institutions.

High-resolution images accompany catalogue entries, enabling close examination of brushwork, surface texture, and condition details that supports connoisseurship and technical analysis. Download options facilitate scholarly use in publications, educational presentations, and research databases;

Creative Commons licensing clarifies permitted uses and attribution requirements. This liberal access policy recognizes that broad circulation enhances rather than diminishes scholarly value, supporting citation, discussion, and critical engagement.

Nordic Art Intelligence's coordination role shows how organizations with regional infrastructure aim to support individual projects facing resource constraints that prevent institutions from independently undertaking catalogue raisonné initiatives. Shared technical platforms reduce development costs through infrastructure pooling, rather than attempting duplicative custom development for each project. Standardized metadata schemas enable interoperability and possibilities for future aggregation. Guidance for best practices drawn from completed projects (such as af Klint and Munch) informs emerging initiatives, preventing common pitfalls and establishing quality standards. Knowledge exchange through regional conferences, working groups, and online forums builds a community of practice supporting mutual learning.

This collaborative infrastructure model distributes sustainability responsibilities across multiple institutions rather than concentrating risk in a single organization vulnerable to funding fluctuations and strategic priority changes. Shared technical infrastructure and knowledge sharing strengthen the broader ecosystem supporting catalogue raisonné development throughout the Nordic region, while individual projects maintain scholarly autonomy regarding content, interpretation, and attribution decisions.

3.4. Ernest Mancoba: Transnational Documentation and Phased Development

The Ernest Mancoba (1904–2002) catalogue raisonné, coordinated by the Ferlov Mancoba Foundation (2025) in Denmark, shows how a digital infrastructure can document transnational artistic legacies over multiple continents, languages, and cultural contexts. The work of Ernest Mancoba, a modernist painter who spent time living in South Africa, Paris, and Denmark, necessitated creating documentation through coordinated research across dispersed archives, collections, and institutional contexts.

Born in South Africa and studying at Fort Hare University in South Africa (historically the first higher education institution for Black Africans), Mancoba emerged within intellectual networks that questioned colonial hierarchies through education and artistic expression. His 1938 move to Paris, where he studied at Académie de la Grande Chaumière and exhibited alongside European avant-garde contemporaries, placed him within European modernist circles. World War II internment and subsequent settlement in Denmark after 1947 added further geographic layers to his artistic biography, with works distributed across three continents and many public and private collections.

The digital catalogue (Ferlov Mancoba Foundation 2025) represents a born-digital approach prioritizing accessibility and collaborative research over traditional print publication and currently documents works in public and quasipublic collections including Kunsten Museum of Modern Art, Aalborg; Statens Museum for Kunst, Copenhagen; and Vanderbijlpark Museum, South Africa. As documented on the project's digital platform, the foundation's phased implementation strategy began with public collection documentation before expanding to private holdings, acknowledging practical challenges of locating and documenting privately owned works that are distributed internationally.

3.4.1. Transnational Infrastructure and Research Challenges

Mancoba's transnational career presents distinct infrastructure challenges that require coordinated documentation across languages, legal jurisdictions, and institutional contexts. South African institutional archives document his early career but suffered disruptions during the apartheid era, when many records of Black artists received inadequate preservation. French archives hold materials from his 1938–47 Paris period but dispersal during World War II created documentation voids. Danish archives document his settled career 1947–2002; connections to earlier phases, however, require international archival research.

Provenance research across three continents requires navigating varied documentation practices, legal frameworks, language barriers, and incomplete records from international art markets. The foundation coordinates with South African, French, and Danish institutions to access distributed archival holdings, which are also supported by digital communication technologies that enable remote consultation, though physical archive visits remain necessary.

The digital-first approach supports continuous updates as works are located and documented, web-based collaboration enables international contributors to submit information and images regardless of geographic location, and phased implementation begins with works most readily documented before extending research to challenging private collections. This methodology acknowledges the practical realities of transnational documentation, which require sustained relationship building with various Galleries, Library, Archives and Museums (GLAMs).

3.4.2. Work in Progress Methodology

The interface notation "Please note the catalogue is a work in progress" explicitly acknowledges the evolving nature of the digital catalogue, contrasting with traditional print catalogues that present as definitive and complete. This transparency serves multiple purposes: setting appropriate user expectations regarding comprehensiveness, inviting community contributions of documentation and corrections, and acknowledging that transnational documentation represents sustained process rather than completable project.

The "work in progress" framing aligns with the digital humanities' embrace of iterative development and versioning rather than definitive publication (Bryant 2002). Regular updates add newly documented works, incorporate revised attributions as research progresses, and expand provenance information as additional archival sources are consulted. Version numbering and change logs document catalogue evolution, enabling scholars to reference specific catalogue states in publications while accommodating ongoing development.

This methodology particularly suits artists whose work distribution and documentation present sustained challenges. For established artists with concentrated collection holdings and comprehensive institutional documentation, traditional catalogue raisonné completion may be feasible. For transnational artists like Mancoba with dispersed works, incomplete archival documentation and ongoing discovery of previously unknown pieces, "work in progress" approaches acknowledge research realities while providing authoritative documentation of works that are currently known.

3.4.3. Institutional Stewardship and Foundation Coordination

The Ferlov Mancoba Foundation stewardship provides dedicated organizational commitment that transcends individual scholarly careers. Founded by Mancoba and his wife, the Danish artist Sonja Ferlov, the foundation preserves his artwork and archival materials documenting their intertwined artistic practices. This institutional continuity ensures sustained attention to Mancoba documentation despite changing museum priorities, grant funding cycles, or individual researcher availability.

Foundation coordination offers key advantages. These include institutional memory that preserves documentation methodologies across staff transitions, sustained relationships with collectors and institutions that support ongoing research, dedicated resources for long-term development, and intellectual property control that enables OA policies without publisher contract constraints. However, foundation models also present barriers. Funding depends on donations and grants rather than stable institutional budgets, staff capacity for comprehensive global research is often limited, and without active collaboration strategies foundations risk insularity from broader scholarly communities.

The foundation's collaboration with Danish museums provides institutional partnerships that expand resources and expertise. Statens Museum for Kunst's (SMK) conservation laboratory conducted technical analysis of Mancoba works, documenting materials and techniques to inform authentication decisions; Kunsten Museum's curatorial staff contributed exhibition history documentation and scholarly interpretations. These partnerships distribute work while building invested stakeholder community and GLAM-supported sustained documentation efforts.

While Mancoba's South African origins extend beyond the Nordic geographic region, coordination through a Danish foundation and representation in Nordic public collections positions the project within Nordic DH infrastructure. The artist's residence in Denmark for more than fifty years and Danish institutional holdings justify his inclusion in Nordic infrastructure discussions, showing how Nordic institutions contribute to documenting artists with regional connections even when their artistic significance extends beyond their specific contexts. Similarly, the Danish sculptor Bertel Thorvaldsen

spent more than forty years working in Rome between 1797 and 1838, yet his catalogue raisonné remains housed at the Thorvaldsens Museum in Copenhagen, which supports the idea that institutional stewardship in an artist's country of origin can transcend geographic location of their artistic production (Thorvaldsens Museum, n.d.). This inclusivity reflects Nordic cultural values that emphasize international engagement and transnational artistic networks rather than narrow nationalist frameworks.

3.5. Emerging Projects: Torsten Andersson and Vera Nilsson

Nordic Art Intelligence's coordination extends to emerging initiatives, which illustrates evolving approaches to digital art historical documentation. The Torsten Andersson Catalogue Raisonné (in progress) and Moderna Museet's (2024) research on Vera Nilsson's drawings represent samples of catalogue raisonné developments in Nordic infrastructure and illustrate varied institutional models and documentation strategies for mid-career and historically underrecognized artists.

3.5.1. Torsten Andersson: Foundation-Led Digital Documentation

The Torsten Andersson Foundation, created by the artist in 2007, is compiling a digital catalogue raisonné of paintings from 1952 to 2009, initiated in 2021 with Nordic Art Intelligence coordination (2025). Torsten Andersson (1926–2009), a Swedish modernist painter, worked in methodical series that investigated how painting functions as language through systematic formal and color relationship studies. This working methodology created documentation that requires systematic series organization and clear articulation of relationships between versions. The catalogue raisonné is structured by series, enabling comparative viewing, detailed dating documentation tracking development sequences, and technical analysis notes documenting material changes as series evolved.

The foundation's artist-initiated origins provide distinctive advantages. Andersson organized archival materials, documented working processes, and articulated conceptual frameworks for his series development, yielding gathered foundational documentation that catalogue raisonné researchers must typically reconstruct from scattered sources. This self-documentation reflects that Andersson's systematic artistic approach extended to comprehensive archives and recordkeeping. The foundation inherited Andersson's complete studio archives, including sketches, preparatory studies, exhibition records, and correspondence that documents artistic relationships and exhibition contexts.

According to Nordic Art Intelligence (2025), the development of the digital catalogue proceeds through phased, chronological implementation of the paintings, beginning with works from the early 1950s and progressing toward later career production. This sequential approach enables systematic research while producing public documentation incrementally rather than waiting decades to publish in order to complete preliminary work. Each phased release adds documented works to a publicly accessible database, supporting ongoing scholarly engagement while research continues on later periods.

Nordic Art Intelligence's coordination provides a technical platform, metadata standardization aligned with Getty terminology (Getty Research Institute n.d.) and vocabularies, and connection to broader Nordic catalogue raisonné community. It was hoped that shared infrastructure would reduce the foundation's development costs and ensure professional standards and long-term sustainability through resources pooled for multiple projects. This collaborative and scalable model could be applicable to mid-size projects of artist documentation: for example, artists of scholarly interest that perhaps lack institutional resources comparable to major figures such as Munch or af Klint.

3.5.2. Vera Nilsson: Museum-Led Research and Exhibition Integration

The Moderna Museet's (2024) research project on Vera Nilsson's (1888–1979) drawings, which represents museum-led comprehensive artist documentation initiative that integrates scholarly research with exhibition planning and public programming. The project investigates Nilsson's work with drawing as memory and narrative, examining her distinctive practice, which combines precise observation and imaginative reconstruction of childhood memories, family histories, and domestic spaces.

Nilsson's recognition as a noteworthy Swedish modernist has grown substantially in recent decades, scholars increasingly acknowledging her innovative approaches to narrative drawing and autobiographical representation. The Moderna Museet project responds to this growing interest by systematically documenting the museum's extensive holdings (of more than 800 drawings) while researching additional works in other collections. This research serves as foundation for an exhibition planned for 2027 to comprehensively present Nilsson's oeuvre, showing the capacity of catalogue raisonné research to support multiple institutional objectives beyond pure documentation.

The research methodology combines traditional connoisseurship (stylistic analysis, dating based on formal characteristics) with archival research (locating documentary evidence regarding creation dates, exhibition histories, provenance), technical analysis (materials identification, condition assessment), and interpretive scholarship (thematic analysis, feminist art historical frameworks, memory studies perspectives). This pluralistic approach reflects historical practice toward contemporary art, integrating varied research methodologies rather than relying solely on formalist connoisseurship.

The museum's institutional infrastructure provides significant advantages through the collaboration including collection access that enables sustained examination of physical works, conservation laboratory facilities that support technical analysis, curatorial expertise that interprets works within broader Swedish contexts, and institutional funding that supports staff time dedicated to research.

Digital documentation developed through the research project will likely inform future public access and OA initiatives, though specific digital publication plans remain under development. Museum collection databases typically serve internal collection management rather than functioning as public-facing catalogue raisonné platforms or online OA publications, creating translation challenges to adapt research documentation for public scholarly access. The project advances museum research initiatives' ability to create foundational documentation that can potentially inform future catalogue raisonné publication while serving immediate institutional objectives including exhibition development and collection interpretation.

3.5.3. Infrastructure Implications

Both emerging projects benefit from Nordic Art Intelligence's aims to connect individual initiatives to broader regional infrastructure through shared technical platforms, support knowledge exchange forums, and problem-solve collaboratively regarding common documentation challenges. The diverse institutional models explored here: artist foundation (Andersson), major museum (Nilsson), foundation that coordinates transnational research (Mancoba), and regional coordinator (Gallen-Kallela)—illustrate multiple viable approaches within Nordic infrastructure, each model offering distinct advantages and facing particular challenges in regard to funding sustainability, scholarly authority, technical infrastructure, and long-term maintenance.

A coordinated regional approach enables smaller projects to benefit from infrastructure investments exceeding the capabilities of individual institutions. Standardized metadata schemas developed through the Munch and af Klint projects inform emerging initiatives, preventing duplicative standards development. Technical platforms that support multiple projects distribute costs across collaborative infrastructure rather than requiring each to independently develop comparable functionality. Knowledge exchange through working groups and conferences enables mutual learning, with experienced projects mentoring emerging initiatives and all projects contributing to best practices.

3.6. Ted Stamm: Research Through Design Case Study

The Ted Stamm Catalogue Raisonné, currently in development, documents the American conceptual artist Ted Stamm's (1944–1984) complete body of work (Fox-Jensen 2025, 2026a), providing a pioneering methodological case study framework of research through design (RtD) approaches (Zimmerman and Forlizzi 2014) that are applicable across catalogue raisonné projects. As a multidisciplinary artist working in SoHo during the 1970s and early 1980s, Stamm developed focused series including Woosters, Dodgers, Zephyrs, Chance, Cancel, and Tags, influenced by urban environments, found objects, and participatory-based work.

Stamm's practice extended out of the traditional studio-based painting to include chance-based systems, shaped canvases, mail art, street interventions (Designators), and collaborative projects, documenting diverse media, ephemeral works, and collaborative contexts. The catalogue raisonné employs RtD methodology integrating interdisciplinary approaches that span art history, design methods, DH infrastructure development, accessibility and sustainability considerations.

3.6.1. Nordic Connections and International Circulation

Stamm's work reached Nordic audiences through exhibitions that connect New York's downtown art scene and Scandinavian institutions. In 1976 Stamm participated in the landmark exhibition *New York—Downtown Manhattan: SoHo*, curated by René Block at Akademie der Künste in Berlin during the Berliner Festwochen, which subsequently traveled in selections to the Louisiana Museum in Humlebæk, Denmark.

In April 1980 Stamm presented a solo exhibition at Galerie Wallner in Malmö, Sweden, introducing Swedish audiences to his systematic approaches to painting and shaped canvas formats during a particularly productive period when he was developing his *Designators* street art series in New York alongside studio works. These Nordic exhibitions positioned Stamm within the international discourse surrounding minimalism, conceptual art, and democratization of artistic production through street interventions and alternative exhibition formats, establishing regional connections that inform current catalogue raisonné research.

3.6.2. Research Through Design Methodology

The project applies RtD methodology integrating design-centered frameworks including contextual design (Holtzblatt and Beyer 2022) that takes stakeholder needs into account, user-centered design (Blair-Early and Zender 2008) for interface development, and participatory approaches (Lauder and Salmon 2014) that engage GLAMs. Iterative development cycles test assumptions, gather feedback, refine approaches, and build collaborative relationships essential for comprehensive documentation.

The project collaborates with the Ted Stamm Estate who possess the majority of Stamm holdings and explores museums with works in permanent collections, and documented oral histories from former colleagues and collaborators who provide contextual information regarding working processes and artistic intentions. These stakeholder relationships provide access to works for documentation, archival materials supplementing visual documentation, and scholarly expertise supporting attribution decisions. Participatory methods create "trading zones" (Lauder and Salmon 2014) in which diverse perspectives inform shared understanding rather than imposing a single authoritative interpretation.

The Ted Stamm slide collection at the Smithsonian's Archives of American Art, comprising approximately 6,450 slides documenting works from 1968–86, benefits archival research. This comprehensive visual archive provides key documentation for works whose current locations remain unknown, destroyed works documented through historical photographs, and provenance research to track ownership changes. Digital photography and slide scanning enable high-resolution documentation while preserving fragile original materials. Similar archival recovery of 1970s New York documentation has shown how such collections restore marginalized practices to the art historical record (Fox-Jensen 2026b).

The project addresses accessibility through OA principles to enable broad scholarly engagement, comprehensive metadata following Getty vocabularies supporting discoverability, and attention to sustainability planning encompassing economic models (estate funding, grant support, institutional partnerships), technical infrastructure (platform selection, preservation planning), organizational sustainability (governance structures, succession planning), and environmental considerations (green hosting, optimized file sizes) informed by the precedents of other Nordic projects.

3.7. Comparative Analysis and Infrastructure Implications

These Nordic catalogue raisonné projects reveal diverse approaches to common challenges (Table 1), each responding to specific artists' legacies, institutional capacities, resource availability, and

technological opportunities. Comparative analysis illuminates patterns, trade-offs, and evolving practices that will inform future initiatives throughout the Nordic region.

Table 1
"Nordic Catalogue Raisonné Comparisons" overview of select artist, works, volumes, dates, institutional affiliation and primary collection, and mode of publication.

Artist	Works	Volumes	Dates	Affiliation	Model
Edvard Munch (1863–1944)	1,789	4 print + digital	2001-13	Munch Museum	OA Digital
Hilma af Klint (1862–1944)	~1,600	7 print + digital (VR AR)	2020-22	Foundation	Hybrid
Akseli Gallen-Kallela (1865–1931)	All known	Digital only	2018-ongoing	Regional + Nordic Art Intelligence foundation	OA Digital
Ernest Mancoba (1904–2002)	All known	Digital only	In progress	Foundation	OA Digital
Torsten Andersson (1926–2009)	Painting 1952-2009	Digital in-progress	2021-ongoing	Foundation + Nordic Art Intelligence foundation	Digital
Vera Nilsson (1888–1979)	Drawings	Research + exhibition	2024-27	Museum	Research + exhibition
Ted Stamm (1944–1984)	All known	Research + archive	In progress	Estate	Research

3.7.1. Institutional Models and Resource Implications

These projects illustrate varied institutional models. Museum-based initiatives (Munch, Nilsson) benefit from institutional resources but face competing priorities. Foundation-managed initiatives (af Klint, Andersson, Mancoba) provide dedicated focus and sustained commitment that transcends museum exhibition cycles, nonetheless depending on fundraising success and perhaps limited technical infrastructure. Regionally coordinated projects (Gallen-Kallela, Stamm) distribute resources and responsibilities across multiple partners, showing scalable approaches applicable to projects of mid-size artist documentation while requiring coordination overhead and stakeholder building.

No single model proves universally superior; rather, each suits particular circumstances. Major artists with substantial institutional holdings benefit from museum-based approaches supported by existing infrastructure. Artists with concentrated family, foundation, or estate holdings benefit from dedicated stewardship to ensure sustained attention. Regional artists lacking a single institutional champion benefit from coordinated approaches that pool distributed resources.

3.7.2. Publication Formats and Access Strategies

The progression from print to digital to hybrid formats reflects evolving technological capacities, changing scholarly communication norms, and emerging philosophical commitments to accessibility. Early projects (Munch prints 2001–8; Woll 2001 and 2008). followed traditional print publication before later digital migration. Recent projects (Gallen-Kallela 2024; Mancoba 2025) began as born-

digital initiatives that prioritize accessibility and updateability. Hybrid approaches (af Klint's print plus planned digital plus VR/AR) serve diverse user communities and access contexts, recognizing that pluralistic rather than single authoritative format serves contemporary needs.

OA digital catalogues (Andersson [planned]; Gallen-Kallela, Mancoba, Munch) exemplify democratization principles central to Nordic DH, removing economic and geographic barriers to access. OA, however, requires sustainable funding models to replace revenue traditionally generated through publication sales or subscription models, with museum operating budgets, foundation endowments, or grant funding free public access.

Digital formats enable "living catalogues" that evolve through ongoing research, versioning, and collaborative contributions (Bryant 2002), contrasting with print's static finality. The Gallen-Kallela catalogue's explicit "work in progress" framing and Mancoba's phased implementation acknowledge that digital platforms support continuous versioning rather than definitive completion. This transition from a catalogue raisonné as a finished publication to an ongoing documentation platform fundamentally transforms scholarly practice, enabling real-time incorporation of new research, attribution revisions, and expanded contextual information.

3.7.3. Design Frameworks and Technical Infrastructure

Design frameworks applied across these projects, contextual design principles, user-centered approaches, visual representation standards, database architecture, participatory methods, and sustainability planning, offer practical guidance applicable throughout the Nordic region and beyond. These frameworks, drawn from multiple project experiences, provide reusable infrastructure components that reduce duplicative development efforts.

Technical infrastructure choices impact long-term sustainability. Projects utilizing open-source platforms (preferred by most Nordic initiatives) reduce vendor lock-in risks and support community-based development. Standardized metadata schemas following Getty vocabularies enable interoperability and possibility of aggregation. Regular data exports in preservation formats ensure content survives technological obsolescence. API access enables third-party tool development that extends a catalogue's utility beyond its original interface.

The four-dimensional sustainability framework (Figure 3) addresses technical, economic, organizational, and environmental considerations and provides essential planning infrastructure that is applicable across projects. Technical sustainability requires platform choices that support long-term viability. Economic sustainability demands funding models that support both initial development and ongoing maintenance. Organizational sustainability depends on institutional commitment, governance structures, and knowledge preservation across personnel transitions. Environmental sustainability increasingly requires attention to carbon footprint through green hosting, optimized data efficiency, and renewable energy commitments.

The accessibility framework (Figure 4) organized around visual design, technical access, responsive design, and WCAG compliance ensures inclusive user engagement across diverse abilities, technologies, and contexts. These frameworks, developed through Nordic project experiences, provide reusable infrastructure informing emerging regional initiatives.

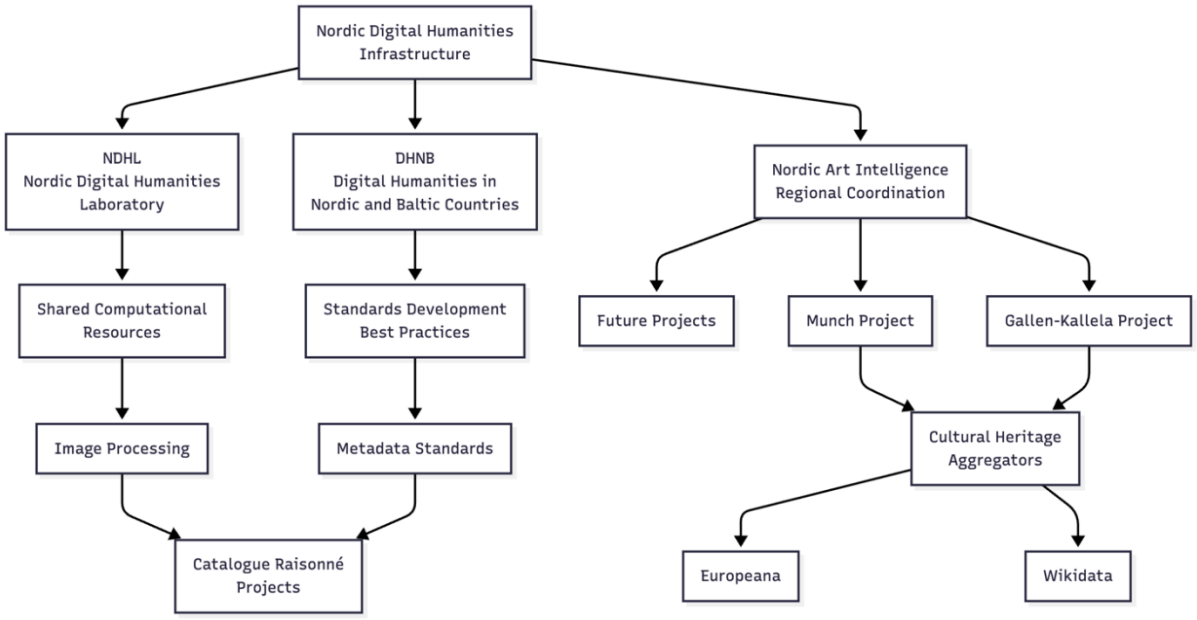


Figure 8: "Nordic Infrastructure Ecosystem and Future Directions" charts relationships between regional organizations (NDHL, NHNB, Nordic Art Intelligence) and catalogue raisonné projects. Created by Fox-Jensen with Mermaid AI, 2025.

3.7.4. Nordic Infrastructure Ecosystem and Future Directions

Nordic catalogue raisonné projects benefit from and contribute to regional DH infrastructure (Figure 8) including DH in the Nordic and Baltic Countries (DHNB 2025) and Nordic DH Laboratory (NDHL 2019–21); other examples could be included such as Norwegian Network for DH and Cultural Organization (DHKO 2025) and Huminfra (Swedish national infrastructure supporting digital and experimental research in the Humanities) coordinating Swedish research infrastructure. These regional organizations provide technical infrastructure, coordinate research initiatives, facilitate knowledge exchange, and connect Nordic scholarship to international DH communities.

Nordic Art Intelligence specifically coordinates catalogue raisonné initiatives, providing technical platforms, project management support, metadata standardization, and community building through conferences and working groups. This specialized coordination role speaks to the particular needs of catalogue raisonné development while connecting to broader DH infrastructure. The coordinated approach enables smaller projects to benefit from infrastructure investments that exceed the capabilities of individual institutions, illustrates successful regional collaboration models that are applicable to other specialized documentation domains, and builds community of practice that supports sustained knowledge exchange and mutual learning.

Recent scholarship also highlights the need for catalogue raisonné projects to critically examine foundational narratives and attribution assumptions. The Hilma af Klint case shows how posthumous mythmaking can obscure collaborative practices and historical complexities (Gregory 2025). As digital platforms enable ongoing revision and versioning, catalogues raisonnés must balance the "living document" concept with the stability required for scholarly citation, particularly when attribution questions emerge that question established market values and institutional buy-in. This tension between historical accuracy and cultural investment brings up the importance of transparent methodology and ongoing scholarship.

The most pressing infrastructure priorities for the Nordic region are coordinated governance through formalized working groups and shared technical platforms for image repositories, metadata aggregation, and preservation services. Metadata standardization must continue to develop to ensure interoperability with cultural heritage aggregators including Europeana (2008) and Wikidata (2012). Environmental sustainability and interface innovation, particularly around VR/AR and AI tools, deserve

sustained attention, provided they do not displace the accessibility commitments that define Nordic DH at its best.

The transformation from "archive to interface" requires attention to dimensions; infrastructure supporting technical functionality, preservation, and interoperability; and thoughtful user interfaces (UI) to widen engagement by diverse user communities. Nordic catalogue raisonné projects show how regional coordination can support individual initiatives while building collective capacity, establishing frameworks and practices that inform digital art historical documentation throughout the Nordic and Baltic regions and contribute to international DH scholarship.

4. Conclusion

This article has examined the digital catalogue raisonné as a critical intersection of infrastructure and interface within Nordic DH, showing how design-centered methodologies advance accessibility and sustainability in digital art historical documentation. Despite these achievements, however, the Nordic region remains underrepresented in catalogue raisonné scholarship relative to its rich artistic and cultural heritage. While major figures such as af Klint and Munch have received comprehensive documentation, numerous Nordic artists including Peder Balke, Carl Larsson, Helene Schjerfbeck, and Anders Zorn lack authoritative catalogues raisonnés. The concentration of resources on internationally recognized artists risks obscuring regionally important practitioners whose work informed national artistic movements and cultural identities.

Increased investment in catalogue raisonné infrastructure, through coordinated funding initiatives, institutional partnerships, and regional coordination mechanisms like Nordic Art Intelligence, could substantially improve Nordic art historical scholarship if funded appropriately, while promoting broader understanding of the region's artistic contributions to international modernism and contemporary practice. As seen by the emerging Torsten Andersson and Vera Nilsson projects, scalable infrastructure models can support mid-career and historically underrecognized artists, yet these initiatives require sustained funding commitments and institutional prioritization to achieve comprehensive coverage.

Through comparative analysis of major Nordic documentation projects such as those regarding Hilma af Klint, Torsten Andersson, Akseli Gallen-Kallela, Ernest Mancoba, Edvard Munch, and Vera Nilsson, this research identifies how interdisciplinary approaches integrating design methods, DH theory, and participatory research address key questions in catalogue raisonné development.

However, an overemphasis on emerging infrastructure and future-oriented projects risks sidelining archival legacies, repeating a broader tendency within DH initiatives to prioritize innovation over sustained engagement with existing historical archival records and analogue objects. The preservation and critical reassessment of earlier documentation remain key for understanding how past curatorial and cataloguing practices inform present day preservation strategies and for safeguarding cultural memory for future generations of curators, art historical scholars and the general public. Digital archives can recover marginalized documentation that earlier cataloguing practices overlooked (Fox-Jensen 2026b).

4.1. Key Findings

Nordic catalogue raisonné projects exemplify diverse institutional models, each with its distinct advantages. Museum-based projects (Munch, Nilsson) are supported by institutional expertise and collection holdings. Foundation-managed initiatives (af Klint, Andersson, Mancoba) provide sustained organizational commitment and family stewardship. Regionally coordinated projects (Gallen-Kallela via Nordic Art Intelligence) distribute resources and responsibilities across multiple partners, illustrating scalable approaches applicable to various artists.

The progression from print to digital to hybrid publication formats reflects evolving technological capacities and philosophical commitments to accessibility. OA digital catalogues (Gallen-Kallela, Mancoba, Munch) exemplify democratization principles central to Nordic DH, while multiformat approaches (af Klint's print, digital, VR/AR) serve diverse user communities and access contexts. Digital formats enable "living catalogues" that evolve through ongoing research and versioning, contrasting with print's static finality.

Applied across these projects, design frameworks, including contextual design, user-centered approaches, visual representation principles, database architecture, participatory methods, and sustainability planning, offer practical guidance that is applicable throughout the Nordic region and beyond. The sustainability framework (Figure 3) addresses technical, economic, organizational, and environmental considerations, providing the key to planning infrastructure, while the accessibility framework (Figure 4) organized around visual design, technical access, responsive design, and WCAG compliance ensures inclusive user engagement.

4.2. Future Directions

The transformation from "archive to interface" requires attention to infrastructure supporting technical functionality, preservation, and interoperability and thoughtful interfaces that enable meaningful engagement by diverse user communities. As Nordic catalogue raisonné projects continue to evolve, the principles articulated here are interdisciplinary collaboration, user-centered design, metadata standardization, sustainability planning, and accessibility prioritization, will remain foundational to create digital resources worthy of the artists they document and the scholarly communities they serve.

Digital catalogues raisonnés represent the achievements of DH scholarship, illustrating how traditional art historical research can be transformed through digital methods while maintaining scholarship and authority. By positioning these projects within broader Nordic infrastructure initiatives including DHNB, DHKO and Huminfra, and Nordic DH Laboratory, this analysis shows how specialized scholarly resources benefit from and contribute to regional DH systems, advancing cultural heritage preservation and democratization of resources throughout the Nordic and Baltic regions.

My own ongoing research on the Ted Stamm Catalogue Raisonné raises questions I have not fully resolved here: specifically, how RtD methodology can be adapted for artists whose careers were transnational and whose archives are split across private estates and public institutions. I intend to test the accessibility and sustainability frameworks developed here against the Stamm project as a case study in a follow-up publication (Fox-Jensen forthcoming).

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